

Education

- 2025–2027 **Master of Statistics [M.Stat]**, *Indian Statistical Institute*, Kolkata, India
Aggregate (till now): 96.0%
 - Currently in 2nd year with specialization in *Theoretical Statistics*.
 - Dissertation Advisors: [Prof. Bodhisattva Sen](#) (Columbia University) and [Prof. Anil K. Ghosh](#) (ISI, Kolkata).
 - Dissertation Title: Variational Inference techniques for State Space Models

2022–2025 **Bachelor of Statistics [B.Stat] (Honours)**, *Indian Statistical Institute*, Kolkata, India
Overall: 92.74% , **Core:** 94.32 %
 - Received First Division with Distinction.
 - Received **Nikhilesh Bhattacharya Memorial Gold Medal** for outstanding performance in Statistics in the B.Stat programme.

2021–2022 **Higher Secondary Education**, *Jenkins School*, Cooch Behar, India
Percentage: 96.4%

Till 2020 **Primary Education**, *St. Mary's High School*, Cooch Behar, India
Class X Marks: 94%

Broad Areas of Interest

- **Machine Learning:** Robust Clustering, Statistical Learning Theory, Stochastic Optimization
- **Sequential Decision Making:** Multi-Armed Bandits, Online Learning, P-Values.
- **Bayesian Statistics:** Variational Inference, Bayesian Nonparametrics, State Space Models

Published papers and Preprints

- 2026 **Stability and Robustness via Regularization: Bandit Inference via Regularized Stochastic Mirror Descent**
B. Halder, I. Sengupta, K. Chowdhury, K. Khamaru.
URL : <https://arxiv.org/abs/2603.10184>
- 2026 **Convex Clustering Redefined: Robust Learning with the Median of Means Estimator**
K. Chowdhury*, S. De*, B. Mandal*, S. Ghosh, S. Das, D. Paul, & S. Chakraborty.
Main Technical Track at **The 40th Annual AAAI Conference on Artificial Intelligence**
URL : <https://doi.org/10.1609/aaai.v40i25.39197>
*Equal contribution

Ongoing Research Projects

- 2026 **Stability and Inference under EXP-3 algorithm**
Supervisor: [Prof. Koulik Khamaru](#), Rutgers University.
Providing an alternate version of EXP-3 algorithm that has best-of-both-worlds optimal regret guarantee and robustness to adversarial corruption.
- 2026 **Evolutionary algorithms for Fair and Robust Clustering**
Supervisor: [Prof. Swagatam Das](#), ISI, Kolkata.
Understanding MOEAs and their application in inducing fairness to well-known clustering algorithms.

Internships and Academic Experiences

- Summer 2025 **Big Data Summer Immersion at Yale (BDSY)**, Yale School of Public Health, Yale University, CT, USA.
Attended a 6-week NIH funded SIBS-program which included classes on Statistical Genetics, Causal Inference and Public Health Modelling Problems. Attended expository and journey lectures of great researchers. Attended Research Groups and conducted collaborative research and presentations under [Prof. Hongyu Zhao](#).
- Summer 2025 Invited for Nurture Camp at IIT Bhubaneswar for qualifying Madhava State Level 2025.
- Spring 2025 Attended **Winter School of Deep Learning** at Indian Statistical Institute, Kolkata
- Winter 2021 Attended **VIJYOSHI Camp** organized by IISc Bangalore.

Honors and Awards

- 2026 **Nikhilesh Bhattacharya Memorial Gold Medal** for the **best performance in Statistics** in B.Stat (Hons.) program (2022 - 2025) at the Indian Statistical Institute
- 2022-2025 Received the prize money for outstanding performance in multiple semesters in the B.Stat program (2022 - 2025) at the Indian Statistical Institute, Kolkata
- 2022 **Kishore Vaigyanik Protsahan Yojana (KVPY)** Fellowship, Department of Science and Technology (DST), Government of India

Invited Talks and Presentations

- January 2026 **AAAI-26 Main Technical Track Poster Presentation, Singapore EXPO**
Title : Convex Clustering Redefined: Robust Learning with the Median of Means Estimator
[\[Poster\]](#)/ [\[Slides\]](#)/ [\[Talk Recording\]](#)
- October, 2025 **D. Basu Memorial Award Presentation at the ISI, Kolkata**
Title : Exploring Bandit Algorithms: A useful tool to deal with Sequential Data [\[Slides\]](#)
- July, 2025 **Symposium on Big Data, Human Health, and Statistics, Yale School of Public Health**
Title : Genetics : The Head and the Heart: Decoding Shared Genetic Architecture Between Coronary Artery Disease (CAD) and Major Depressive Disorder (MDD).

Selected Coursework Projects

- July, 2025 **Studying genetic overlap of Coronary Artery Disease and Major Depressive Disorder using GWAS summary statistics, YSPH, Yale University, CT, USA**
Supervisor: [Prof. Hongyu Zhao](#)
Analyzed large-scale GWAS data (CAD and MDD) to assess shared genetic architecture. Estimated SNP heritability and genetic correlation via LDSC, and identified several novel pleiotropic loci using PLACO and Meta Analysis, supporting partial genetic overlap between CAD and MDD.
[\[PPT\]](#)/[\[Poster\]](#)
- May, 2025 **Temporal Sentiment Analysis – A Statistical Exploration of Societal Distress and Reaction, Indian Statistical Institute, Kolkata, India**
Supervisor: [Prof. Kiranmoy Das](#) and [Prof. Anil K. Ghosh](#)
Modelling momentary anger across time caused due to incidents that triggered an immediate and widespread emotional response throughout the city. Explaining variations in various behavioral responses for various sections of society. Modelling was done using GEE equations and Quasi-Likelihood optimization. Data was collected through surveys across households all over Kolkata, India.
[\[PPT\]](#)/[\[Report\]](#)

May, 2025 **Time Series Analysis of Sector-wise GDP in India using MOSPI Data**, *Indian Statistical Institute*, Kolkata, India

Supervisor: [Prof. Kiranmoy Das](#) and [Prof. Anil K. Ghosh](#)

A comprehensive time series analysis on various sectors of GDP such as Agriculture, Fishing, etc. The analysis includes, using Jump Regression for trend estimation, SARIMA Model fitting and forecasting.

[\[PPT\]](#)/[\[Report\]](#)

May, 2025 **Studying the Structure of the Local Neighborhood of a Randomly Selected Vertex of a Large but Sparse Erdős-Rényi Binomial Random Graph**, *Indian Statistical Institute*, Kolkata, India

Supervisor: [Prof. Antar Bandyopadhyay](#)

Statistically studying the local (d-depth) neighborhood of a randomly selected vertex of a large but sparse Erdős-Rényi random graph through simulations and theory available in the field.

[\[PPT\]](#)/[\[Report\]](#)

November, 2024 **Demographic Analysis and Comparative Study of Maharashtra and Manipur using NFHS-5 dataset**, *Indian Statistical Institute*, Kolkata, India

Supervisor: [Prof. Kajori Banerjee](#)

A study aimed to provide insights into the disparities and developments that shape Maharashtra and Manipur, the socio-economic dynamics, health indicators, and lifestyle patterns that define the populations of these two diverse regions offering an understanding of India's multifaceted socio-economic landscape. Key metrics will include income and expenditure, education levels, and reproductive health.

[\[PPT\]](#)/[\[Report\]](#)

Other Achievements and Scholarships

2026 Joint Admission Test for M.Sc (JAM) Mathematical Statistics : AIR 1

2026 Graduate Aptitude Test in Engineering (GATE) Statistics : AIR 2

2025 Simon Marais Mathematics Competition East-Division Qualification in Pair category

2025 Madhava Contest State-Level Qualification and invitation for Nurture Camp

2023 Madhava Contest State-Level Qualification

2022 JEE Mains: AIR: 369 (99.967 percentile)

2022 JEE Advanced: AIR 808

2022 WBJEE: Rank 5

2022 ISI Entrance Test: Rank 17 with scholarship

2021–2022 KVPY Fellowship (SA AIR: 443; SX AIR: 437)

2021 Qualified NTSE Stage-I (Rank: 43) and JBNSTS with scholarship

Other Information

Programming Python, R, C, MATLAB

Document Preparation LaTeX, MS-Office, Google Workspace

Mathematical Techniques Statistical Modeling, Optimization, Graph Theory

Hobbies Photography, Cricket, Badminton, Painting

The information provided is correct to the best of my knowledge.